

Multi-color LED Marker for Dynamic Target Tracking in Wide Area

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ABSTRACT

To present augmented reality in dynamic scenes like sports, a wide tracking range is required as well as accuracy and speed. Conventional tracking markers had a complicated shape, which resulted in being weak against image blur and low image resolution, leading to a narrow tracking range. This paper proposes multi-color LED markers for tracking a dynamic object in a wide area. They can emit consistent light and express a unique ID by a few color pixels, which are efficiently extracted using a camera's short exposure time. This marker was tested at various distances and fast rotation, where the high identification accuracy was evaluated.

Index Terms: Human-centered computing—Interaction paradigms—Mixed / augmented reality; Computing methodologies—Computer vision—Tracking

1 INTRODUCTION

Fast and accurate tracking for widely moving objects is required to present augmented reality (AR) in dynamic situations like sports. In such a situation, putting tracking markers onto the target surface is suitable in terms of accuracy and speed, and their placement should be designed to be easily recognized even in low resolution.

LEDs are focused as markers in this paper as they can consistently emit strong light in the same color range. It should be noted that their required power is small, which is not a problem especially in some cases, like the combination with a drone. The visible colorful LEDs may limit AR image superimposition onto the target object, but this will be improved by the future usage of infrared (IR) lights of multiple wavelengths and an IR-color camera which detects different IR wavelengths [2].

Many previous methods using LED markers have taken blinking ways [4], which generally needs synchronization with the camera and is not suitable for wide tracking. The multi-color-coding way is considered in this paper, which has been explored only in the printed colors and is weak against changing lighting conditions [3].

We propose multi-color LED markers for wide-area target tracking. LEDs have the advantage of stable color and luminance under any lighting condition compared to traditional color-printed markers. The marker shape is simple but can express unique IDs using a few color pixels. This marker also has an advantage in image processing; only the bright areas of a marker can be extracted by limiting camera exposure time, which is also effective for eliminating motion blur in fast movement. For posture identification, the color placement on each LED is considered so that the combination of adjacent marker colors can express unique IDs. Note that each color is always static; IDs are recognized just from the relationship of adjacent marker colors.

As a first step, this paper focuses on a spherical cage made of wire frame as a target object, which mounts a drone and autonomously moves around the wide field, used in a new e-sports

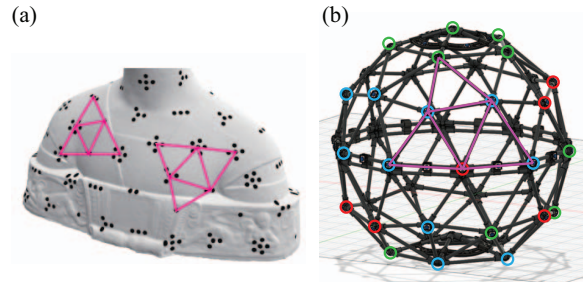


Figure 1: (a) Extended dot cluster marker, cited by Figure 2 in [5]. (b) Our multi-color LED markers whose placement refers to (a).

game combined with AR [1]. This paper discusses the proper arrangement of different colors and ID-recognition algorithm with a reference to previous research [5], and evaluates if our marker works in a wide range of distances and fast rotation.

2 METHOD

The placement of different colors and the ID-recognition algorithm refer to a previous study of extended dot cluster marker (EDCM) [5], as shown in Figure 1 (a). This previous marker consisted of different numbers of dots, and its unique ID could be recognized using adjacent four triangles providing six markers. We replace this dot cluster with a multi-color LED.

One different factor in our study compared to the previous one [5] is, that our adjacent markers can have the same color. There are two reasons for it. One reason is that if a part of the LED markers is occluded, the color does not greatly change and the tracking performance is not affected; the previous method may be susceptible to this because if a part of a cluster marker is occluded by hands and fingers, the recognized number of dots within that marker might be reduced. The second reason is that, since some adjacent markers would have the same color, the number of different colors can be reduced, which leads to enhancing the color-extraction accuracy.

In this study, we target a cage made of triangle shapes for e-sports [1] as shown in Figure 1 (b). By brute-force search to find a unique arrangement of markers, it was revealed that our LED marker can work with only three different colors. This is owing to the sparse wire frame of the target cage; if a large number of LEDs are put onto the target object, the necessary number of different colors would increase.

Note that, our method is also different from the previous one in the marker placement; the number of adjacent markers for one marker is not always six, sometimes five, while it was always six in the previous method. This allows keeping the usage of four adjacent triangles while loosening the restriction of marker placements.

As the other novel point, our methods require setting the camera's exposure time to be short to detect only the brightest points and eliminate motion blur even in the fast object movement.

3 EVALUATION

3.1 Evaluation environment

The picture of the evaluation environment is shown in Figure 2.

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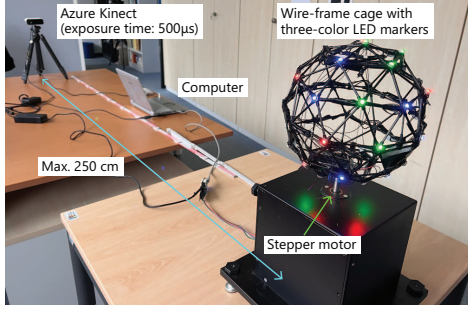


Figure 2: Evaluation apparatus.

The multicolor LEDs (WS2812B, BTF-LIGHTING), each color is programmable, were put onto a cage that was made of black triangle wire frames assuming the usage in the new e-sports [1]; the wire frame is sparsely placed. The three colors of LEDs were red, green, and blue. The cage diameter was approx. 23 cm. A stepper motor (Nema 17, OSM Technology Co.,Ltd.) with a controller (Stepper Brick DRV8811, Tinkerforge GmbH) was used for automatic rotation. The used notebook computer was Panasonic Let's note CF-SV (Intel® Core™ i7-8565U).

The used camera was Azure Kinect (Microsoft, 1280×720 px, 30 fps, color, field-of-view (FOV): W90°×H59°), whose exposure time was set to 500 µs. The cage was fixed on the stepper motor, and the distance between the camera and the center of the cage varied at 0.5, 1.0, 1.5, 2.0, and 2.5 m. The height of the camera and the cage was always the same.

Since the FOV of the used camera is too wide, it is helpful to consider the conversion of distance depending on the camera's FOV. Let the cage radius be r (=11.5 cm) and the cage distance from the principal point of the camera be d . By making the cage occupy the same number of image pixels at any distance, the converted distance d' at different FOV ($= fov'$) from original distance d at original FOV ($= fov$) can be formulated as

$$d' = \frac{r}{\tan(fov' * \theta / 2)} \quad (1)$$

$$\text{where } \theta = \frac{2 \arctan(r/d)}{fov}. \quad (2)$$

For example, assuming that $fov = 59^\circ$, the vertical FOV of the used camera, the success rate at $d = 2.0$ m in our system would be converted into $(d', fov') = (3.937 \text{ m}, 30^\circ)$ and $(11.81 \text{ m}, 10^\circ)$, which is said to be a sufficient range of distance.

3.2 Evaluation of tracking

The cage rotated at 9° steps all around, and five frames were captured by the camera for each rotation, resulting in 200 frames in total. The ID recognition was conducted using each frame, whose result was directly drawn onto the image and used for the manual judgement of tracking success by the corresponding author. The success rate was calculated as the average rate for all the frames.

The evaluation result is shown in Table 1. The success rate was almost 100% at 0.5-2.0 m, and it dropped to 83.0 % at 2.5 m. The examples of captured and result-drawn images at 0.5 and 2.5 m are shown in Figure 3 (a) and (b). The cage diameter in the image at 0.5 m distance was 330 px, much smaller than the image height (720 px), which implies the possibility of tracking at small distances with a small FOV. That at 2.5 m was 60 px; the success rate over 80 % is substantial even with such a small image size.

The fast rotation at 300 RPM was also tested, and its appearances from both iPhone 13 Pro (30 fps, the estimated exposure time: around 30 ms) and Azure Kinect are shown in Figure 3 (c). The

Table 1: The success rate of ID recognition of multi-color LED markers at various distances

Distance [m]	0.5	1.0	1.5	2.0	2.5
Success rate [%]	100.0	100.0	99.0	99.0	83.0

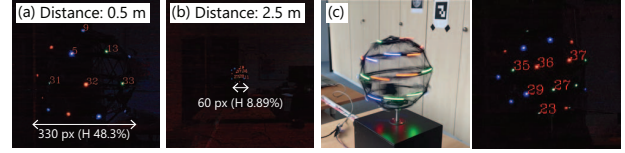


Figure 3: (a,b) Example of captured images using Azure Kinect at 50 and 250 cm, respectively. (c) Captured images of fast rotation (300 RPM); left shows a picture taken by iPhone 13 Pro at 30 fps while right shows the image taken by Azure Kinect with short exposure time at 100 cm distance.

latter has no motion blur thanks to short exposure time while the former has it.

The processing time was calculated as 4.950 ms on average with a standard deviation of 1.811 ms using a 1000-frame recorded video while the cage rotates at $22.5^\circ/s$. This means almost 200 fps tracking is possible, which is sufficient for normal AR superimposition.

4 DISCUSSION AND CONCLUSION

It is notable that the bright colorful LEDs work as tracking markers and a camera of limited exposure time facilitates their easy detection and ID recognition. This method is also robust against specular reflection since LEDs are brighter than it. Sunlight may obstruct them because of similar luminance, hence indoor usage is preferable.

The cause of false detection at a distance is estimated due to mistriangulation, which is owing to missing LEDs caused by a lack of LED luminance or too short exposure time. However, this would be improved by using the camera of a narrow FOV, as discussed in Section 3.1 and 3.2.

In conclusion, our multi-color LED markers have been evaluated to have both high detection accuracy and speed even in a wide area. Our markers would be suitable for usage in dynamic situations such as sports, if the requirement for electricity is given. In the future, it is expected that our method would be applied in the actual application of highly dynamic scenes like sports.

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